



ELE101

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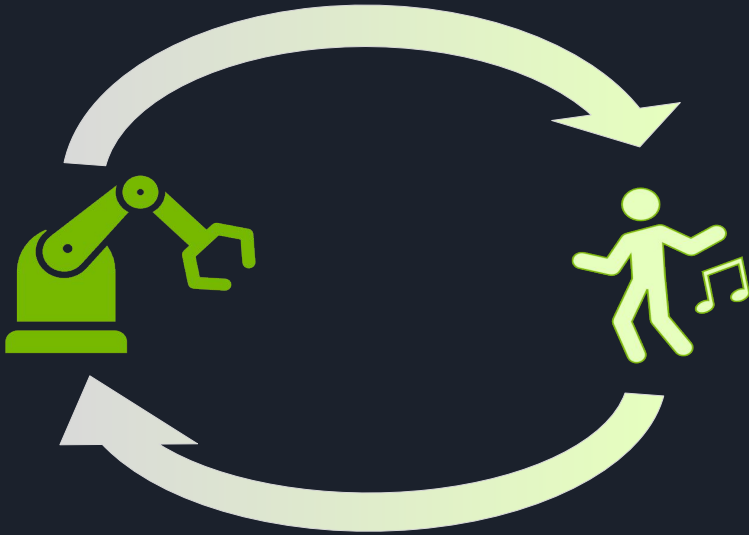


Outline

- AI Introduction
- NVIDIA DLI
- ELE495

Human Vs Machine Learning

Relaxed Alertness



Human	Machine
Rest and Digest	Training
Fight-or-flight	Prediction

Beginning of Artificial Intelligence

Computers are made
in part to complete
human tasks



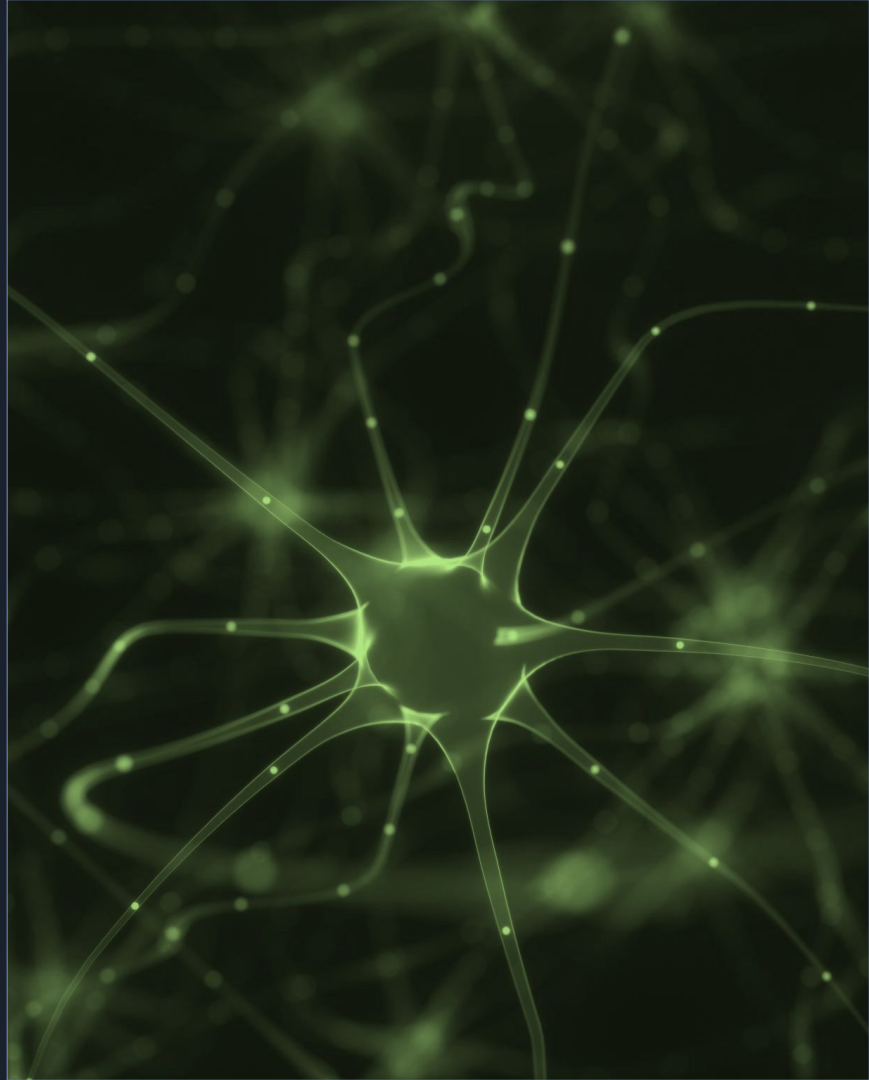
Early on, generalized
intelligence looked
possible



Turned out to be
harder than expected

Early Neural Networks

- Inspired by biology
- Created in the 1950's
- Outclassed by Von Neumann Architecture



Expert Systems



Highly complex



Programmed by hundreds of engineers



Rigorous programming of many rules

Expert Systems - Limitations

What are these three images?





How Do Children Learn?

- Expose them to lots of data
- Give them the “correct answer”
- They will pick up the important patterns on their own

The background of the slide features a series of overlapping, wavy, ribbon-like shapes in shades of light green and yellow, creating a sense of depth and movement. The shapes curve from the bottom left towards the top right.

The Deep Learning Revolution

Data

- Networks need a lot of information to learn from
- The digital era and the internet has supplied that data



Computing Power

Need a way for our artificial “brain” to observe lots of data within a practical amount of time.

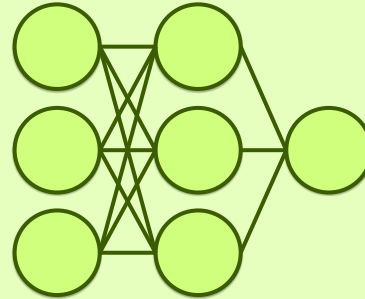


The Importance of the GPU

A Rendered Image

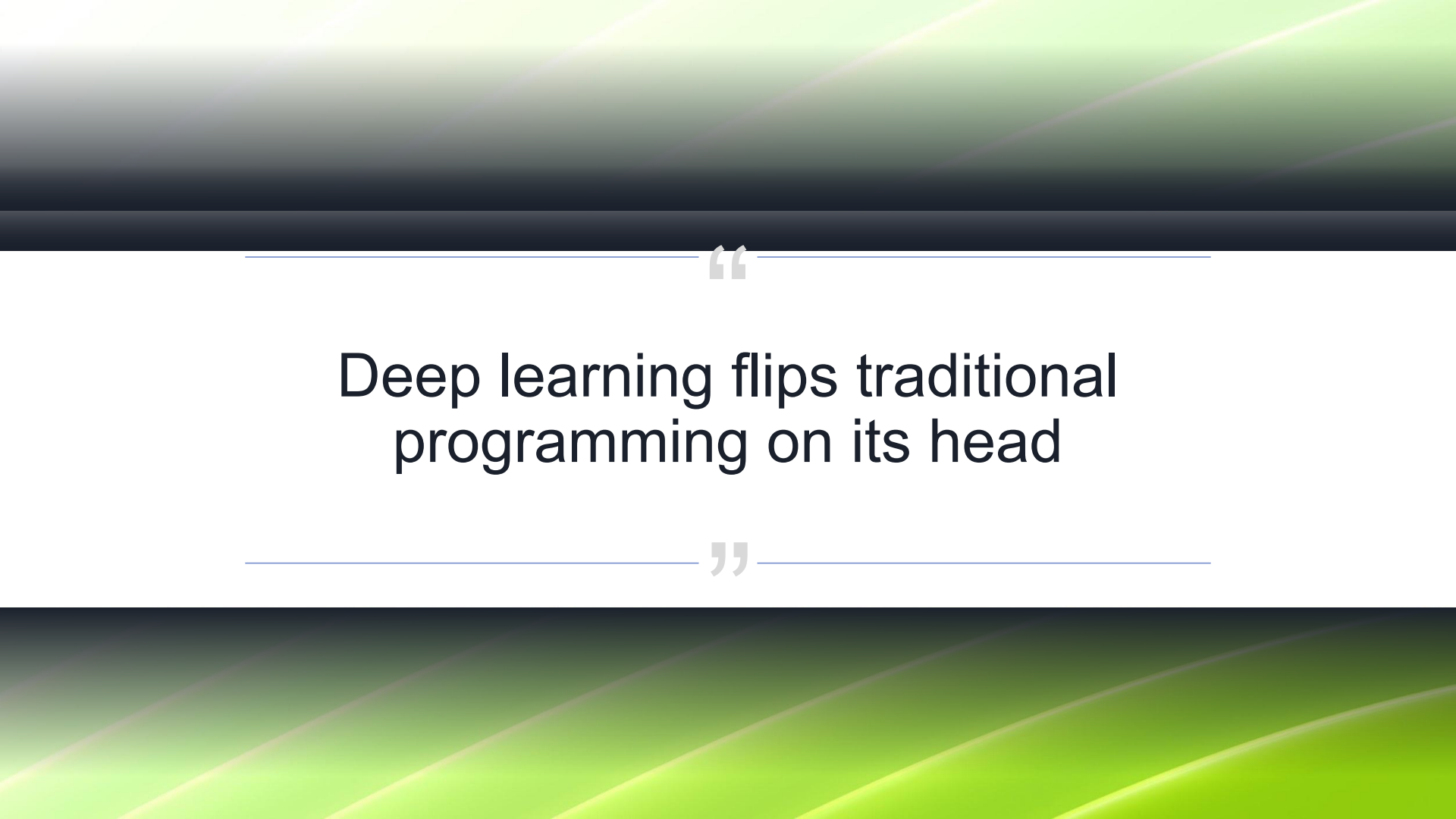


A Neural Network



What is Deep Learning?

The background of the slide features a series of overlapping, wavy, ribbon-like shapes in shades of light green and yellow, creating a sense of depth and movement. The shapes curve from the bottom left towards the top right, with some areas appearing more saturated than others, giving it a 3D effect.



“

Deep learning flips traditional
programming on its head

”

Traditional Programming

Building a Classifier

1

Define a set of rules for classification

2

Program those rules into the computer

3

Feed it examples, and the program
uses the rules to classify

Machine Learning

Building a Classifier

1

Show model the examples with the answer of how to classify

2

Model takes guesses, we tell it if it's right or not

3

Model learns to correctly categorize as it's training. The system learns the rules on its own

The background of the slide features a series of overlapping, wavy, ribbon-like shapes in various shades of green and yellow, creating a sense of depth and movement. The shapes are layered, with some appearing to be in front of others, and they curve across the frame from the bottom left towards the top right. The colors transition from a pale yellow-green on the left to a more vibrant green on the right.

This is a Fundamental Shift

When to Choose Deep Learning

Classic Programming

If rules are clear and straightforward, often better to program it

Deep Learning

If rules are nuanced, complex, difficult to discern, use deep learning

Deep Learning Compared to Other AI

Depth and complexity of networks

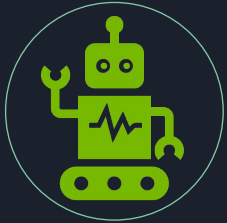
Up to billions of parameters (and growing)

Many layers in a model

Important for learning complex rules

How Deep Learning is Transforming the World

Computer Vision



ROBOTICS AND
MANUFACTURING



OBJECT
DETECTION



SELF-DRIVING
CARS

Natural Language Processing



REAL-TIME
TRANSLATION



VOICE
RECOGNITION



VIRTUAL
ASSISTANTS

Recommender Systems



CONTENT
CURATION



TARGETED
ADVERTISING



SHOPPING

Reinforcement Learning



ALPHAGO
BEATS WORLD
CHAMPION IN
GO



AI BOTS BEAT
PROFESSIONAL
VIDEOGAMERS



STOCK TRADING
ROBOTS

Image Classification

Objective

- Recognize the main subject in the image and assign a label to the image



Datasets for Image Classification

Architectures

- LeNet
- AlexNet
- VGG-16
- Inception V1
- ResNet-50
- Xception
- MobileNet



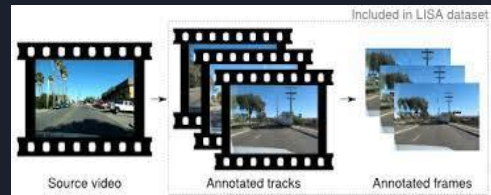
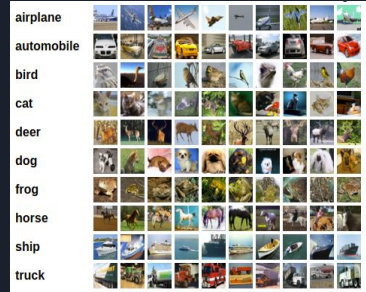
Object Detection

Objective

- Predict the class and the position of multiple objects in the image



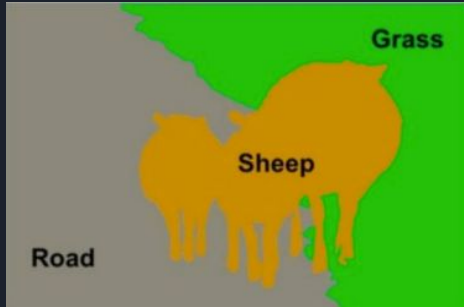
- Sliding Window
- R-CNN
- Fast R-CNN
- Faster R-CNN
- YOLO
- SSD
- Transformers



Semantic Segmentation

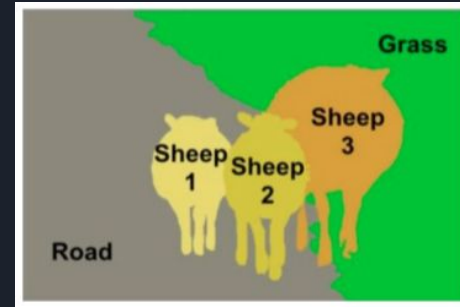
Semantic Segmentation Objective:

- Classify all pixels
- Don't differentiate instances.



Instance Segmentation Objective:

- Classify pixels but differentiate instances (predict segmentation mask)



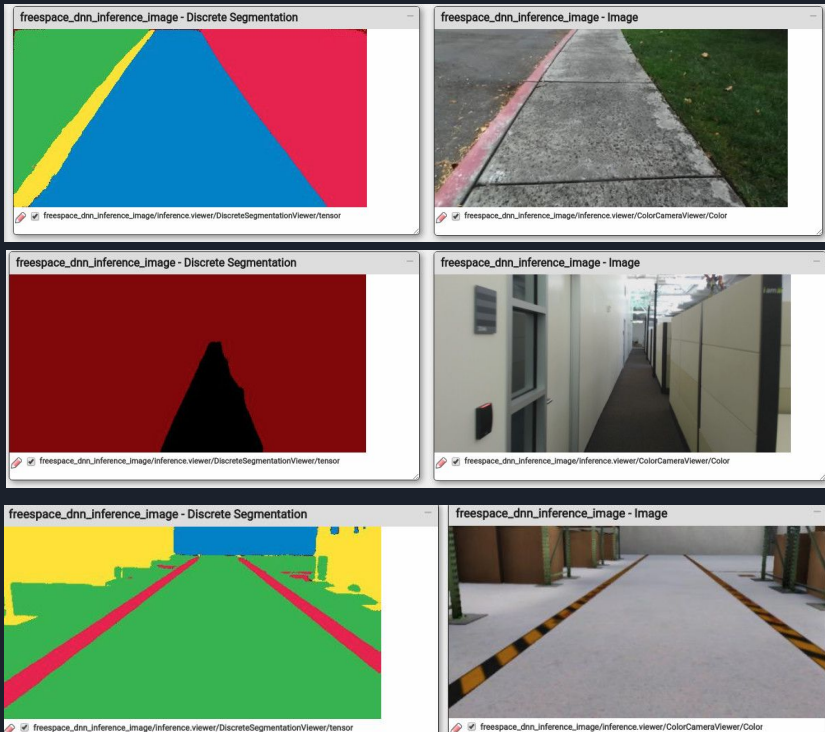
Datasets for Semantic Segmentation

Architectures

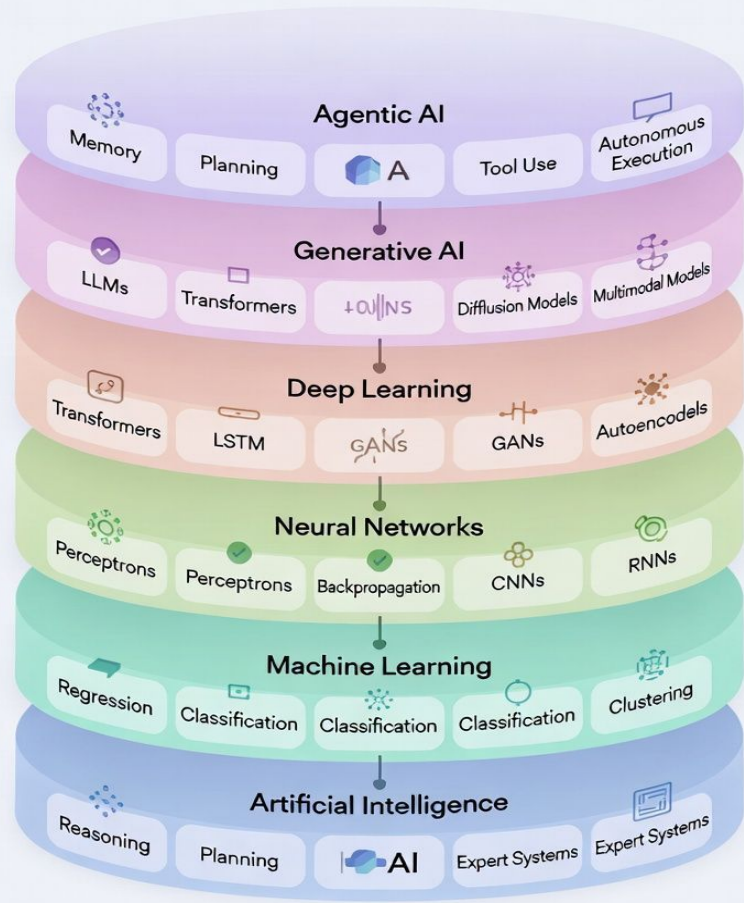
- Fully Convolutional Networks
- SegNet
- U-Net
- RefineNet
- Learning Deconvolution Network
- Mark R-CNN



Applications: Free Space Detection



Layers of AI



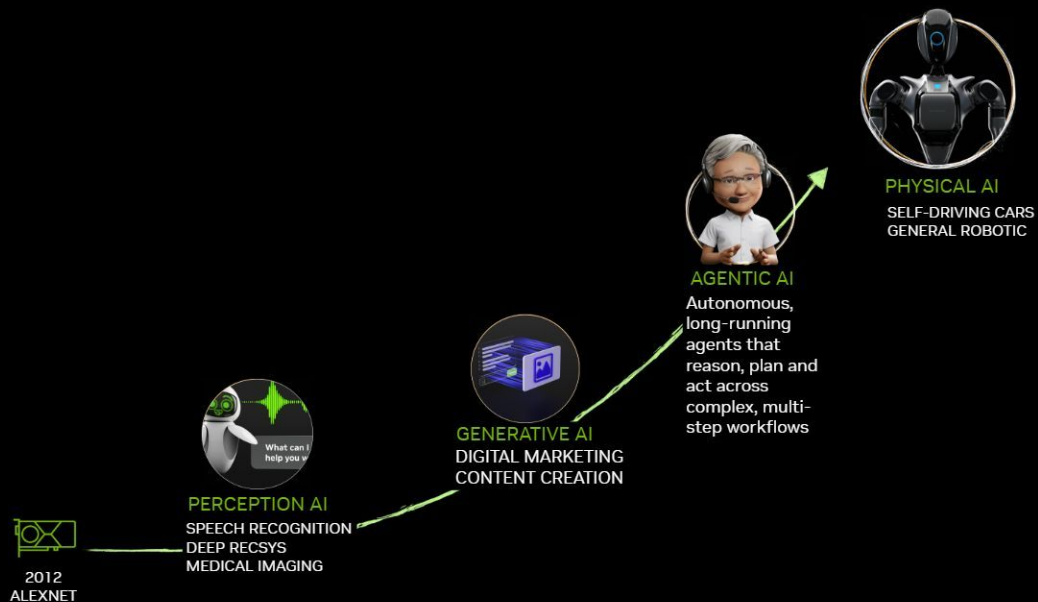
AI Evolution

- Machine learning didn't disappear when deep learning arrived.
- Neural networks didn't become irrelevant when transformers became popular.
- We often talk about the top layer first: agents, tools, automation, systems that appear to “think” or act on their own.
- But those ideas only make sense if you understand what sits underneath them.

Agentic AI

Evolution of AI

Agentic AI Enables More Powerful AI Applications



From Planning to Acting



PERCEIVE

Reads your research topic, existing files, and project structure



REASON

Plans file structure, content outline, and folder layout



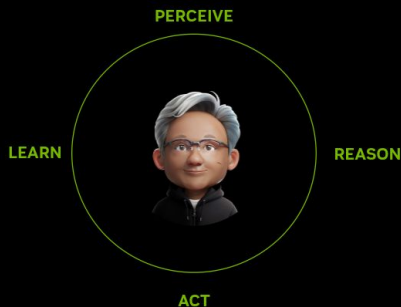
ACT

Calls file_writer — creates dirs, writes files



LEARN

Refines the plan based on what it successfully wrote



Jetson AI Fundamentals

Getting Started
with AI
on Jetson Nano



System Setup



Camera Capture



Classification



Regression

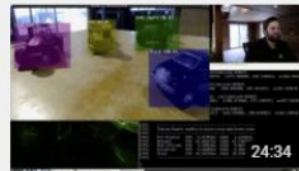
Hello AI World



Image Recognition



Object Detection



Data Collection + Training

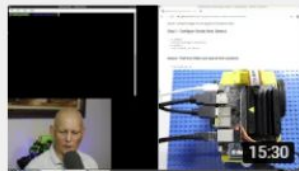


Semantic Segmentation

JetBot



Hardware Setup



Software Setup



Collision Avoidance



Road Following



ELE495

- Capstone Project
- 2223-Yaz: JetBot Material Classification
 - <https://youtu.be/4ZHhM5hg72c>
- 2324-Bahar: Conveyor Belt System, Robot Arm Project
- 2324-Yaz: JetBot Autonomous Parking
 - <https://github.com/orgs/ELE495-2324Summer/repositories>
- 2425-Bahar: Autonomous RF Emitter Location Finding
 - <https://github.com/orgs/ELE495-2425Spring/repositories>
 - Grup alfa: https://www.youtube.com/watch?v=lptra_yDA2u8
- 2425-Yaz: Voice Controlled Autonomous Vehicle
 - <https://github.com/orgs/ELE495-2425Summer/repositories>
- 2526-Bahar: Production with Robot Arm
 - <https://github.com/orgs/ELE495-2526Spring/repositories>



Survey

- Take the survey before you go!





Thanks!

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